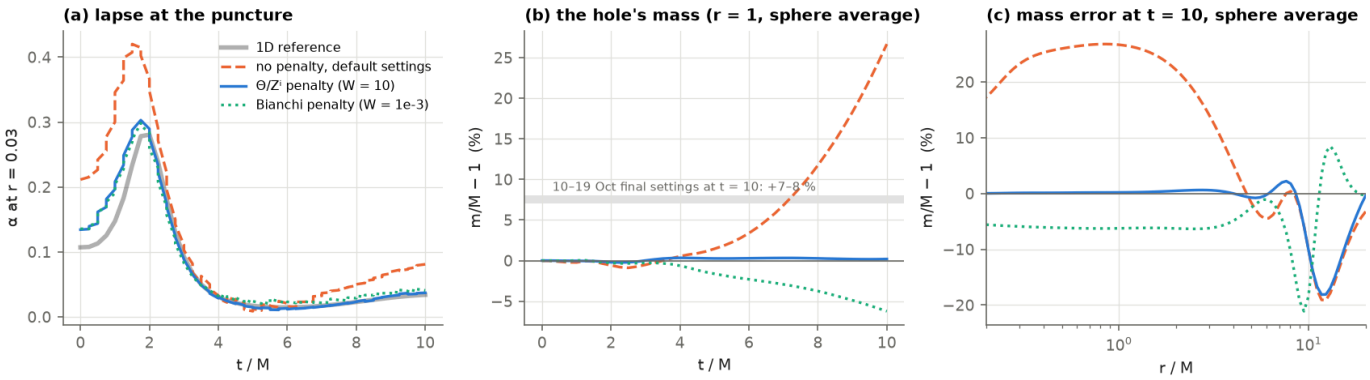
Black-hole puncture CCZ4 PINN: constraints, mass and drift

Work of 20–26 October 2026 · O. Reula, with Claude · same code and problem as the 10–19 October summary (J. Massó): one 6×40 network per 0.25 M window, single puncture m = 0.5, t ≤ 10 M, SSBroyden throughout. All results graded against the collaborators' 1D spherical reference (dr = 0.005) and the Misner–Sharp mass, not the training loss. **Draft**: the control run that attributes the last finding is running.

Bottom line. The trained networks are not Ricci-flat near the horizon, and the residual loss cannot see why: a constraint-violating solution still satisfies the CCZ4 evolution equations. Adding the CCZ4 constraints Θ and Z^i (zero for a true solution) to the loss fixes the black hole: to t = 10 M its mass stays within **0.35 %** (final settings without it: +7–8 %), the lapse at the puncture within 0.003 of the reference, and the lapse error near the hole drops 10×. What the penalty does not change is a mass **deficit** that leaves the hole at t ≈ 2 and travels outward with the gauge pulse, growing to about −20 % (sphere average) at r ≈ 10–12 by t = 10. It is the same with or without the penalty. Both penalised runs also hold the puncture in place: drift 0.010 and 0.007 by t = 10, against 0.13 without a penalty and 0.023 for the 10–19 Oct final settings. Point-symmetric sampling clouds stop it too (25× less).



Overview, t ≤ 10 M. (a) α at r = 0.03 against the reference. (b) Misner–Sharp m/M − 1 at r = 1, outside the horizon (r ≈ 0.25), at each window end; the band is the 10–19 Oct final settings at t = 10. (c) m/M − 1 against r at t = 10. Mass errors are averaged over spheres (48 directions) about the tracked puncture. Runs: `single_a` (code defaults, the 10–19 Oct "first baseline", no penalty); `long_con10` and `long_bia3` (final settings, i.e. 48k core points and 24k QN steps per window, plus a penalty, section 4).

1. Collocation points

Every window (0.25 M) draws a fresh set of points every 250 QN iterations: 96 times per window with the final settings (24k iterations), 48 with the defaults (12k). Per draw, with the final settings:

set	points	distribution in space
interior, core cloud	48 000	Gaussian, σ = 0.5, about the tracked puncture (final settings only)
interior, focus cloud	48 000	Gaussian, σ = 2, about the tracked puncture (60 % of the 80 000 base points)
interior, bulk	32 000	uniform in the ball r ≤ 24 (the other 40 %)
initial surface (t = t_{start}) and end surface (t = t_{end})	12 800 each	8000 as focus + bulk (60/40), plus 4800 from the core Gaussian
outer boundary (Sommerfeld)	16 000	on the sphere r = 24

No point is drawn within 0.02 of the puncture. Interior times follow a truncated exponential from the window start, widened to uniform over the window by 70 % of the draws. The Θ/Z^i penalty uses all 128 000 interior points; the Bianchi term uses the 4096 last ones (core cloud). By radius: 29 % of the interior points lie at r < 1, 43 % at 1–5, **10 % at 5–15** (24 % of the volume, where the outgoing mass deficit of section 5 travels) and 19 % at 15–24. The defaults run has no core cloud: 80 000 interior points and 8000 per surface.

2. The loss is not the accuracy; the reference is

With the final settings the code reproduces the 10–19 Oct numbers against the reference: α error at the puncture **0.022 / 0.032** at t = 0.25 / 1 (theirs 0.019 / 0.024), 5× better than the defaults (0.108 / 0.153). On fresh points never trained on, the loss is 5–10× higher than the value logged in training, and the logged loss *falls* (1e-6 → 2e-7) over a stretch where the mass error of the same run grows to +28 %. Every result below is therefore judged by the reference and by the mass, which in vacuum spherical symmetry must equal M at every radius and in any slicing (checked exact to 4e-15 on the initial data).

3. Where the mass error comes from

Each window's network is evaluated at its start and end, so the error at any radius splits exactly into the initial-data fit, the growth *inside* windows, and the jumps at the hand-offs between windows. In `single_a` at t = 10 (r = 1): initial −1e-4, inside windows +0.30, hand-offs −0.015. So the error is **built inside the windows**, not inherited at the hand-offs, and its radial density is positive only at r ≲ 0.3, around the horizon; the integral carries it outward.

The cause is direct. We rebuilt the 4D metric and the Z4 vector Z_μ from the network's CCZ4 fields (Alic et al. mapping; exact Schwarzschild passes it to 7e-15) and evaluated the Z4 field equations $E_{\mu\nu} = R_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu$. In window 1, the rms of E is 0.2 at $r \in [0.3, 0.7)$, 7e-3 at [0.7, 1.5) and 4e-4 at [3, 6): **the network's spacetime is not Ricci-flat at the ~5 % level around the horizon**, while the CCZ4 residual there is 2.4e-4. The residual loss sees the evolution equations, and these admit solutions with Θ, $Z^i \neq 0$, which carry wrong mass. Only the initial data rule them out.

4. Constraint penalties

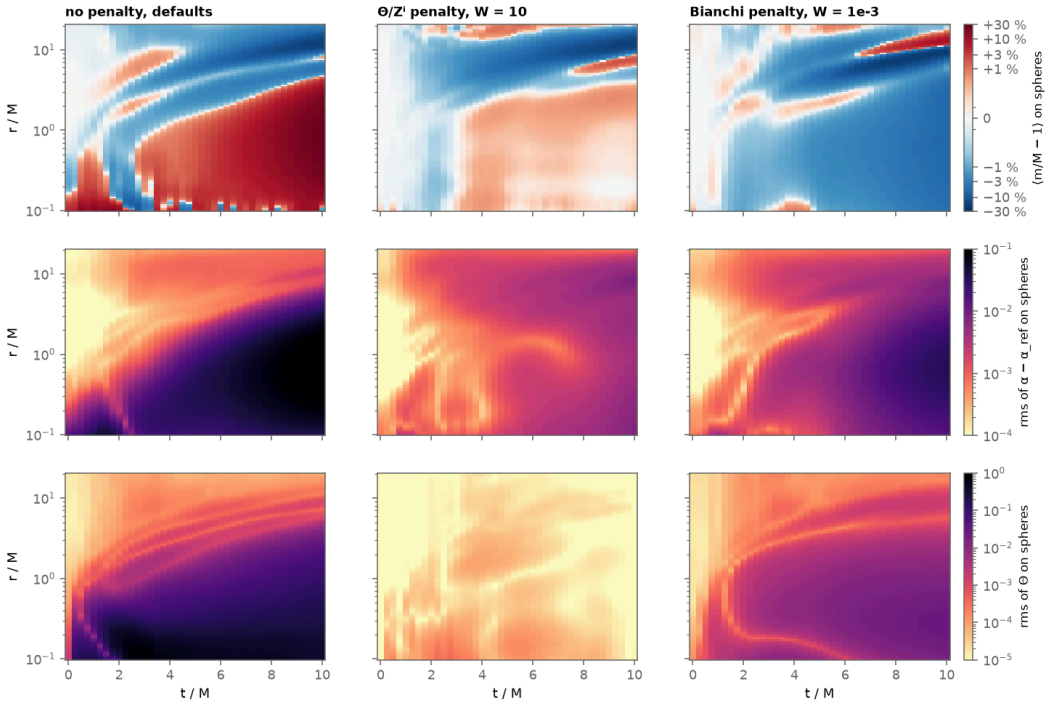
Two ways to rule them out, each a term added to the loss, zero on the true solution:

- **Direct** (`--w_constraint W`): W·mean($\Theta^2 + |Z^i|^2$) over the interior points. Cheap.
- **Contracted Bianchi** (`--w_bianchi W`): for any metric and covector, $\nabla^\mu(E_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}E) = \Box Z_\nu + R_{\nu\mu}Z^\mu$. Z obeys a linear wave equation sourced only by the divergence of the field-equation residual, which is the part of the error that feeds the constraint modes. The term W·mean(B^2), $B = \Box Z + R\cdot Z$, on 4096 core points, checked to 5e-16 on random metrics. It needs third derivatives of the network and costs ~35× the residual per point.

At t = 2 (8 windows), both cut the core errors ~10× against the same run without them: core α error 1.2e-2 → 7e-4 (direct) / 1.1e-3 (Bianchi), max Θ 0.45 → 3e-4 / 1.4e-2. Continued to t = 10:

t = 10	no penalty, defaults	direct, W = 10	Bianchi, W = 1e-3	10–19 Oct final
α at r = 0.03 (reference 0.034)	0.081	**0.037**	0.041	0.041
α error, core 0.05 ≤ r < 0.5 / mid 0.5 ≤ r < 5	6.6e-2 / 1.0e-1	**6.0e-3 / 3.5e-3**	2.2e-2 / 2.7e-2	–
max |Θ|	0.42	**3e-5**	2.9e-2	–
mass of the hole, m/M − 1 at r = 1	+27 %	**+0.19 %**	−6.2 %	+7–8 %
worst outer deficit (radius)	−19 % (12.4)	−18 % (11.7)	−21 % (9.5)	≤ 17 % (4–12)
puncture drift $\|x_p\|$	0.13	0.010	0.007	0.023

The direct term is the clear choice for the hole: Θ falls 10^4×, and the lapse and the mass at the puncture match the reference. The Bianchi term, the more physical idea, does less at the weight tried and makes the hole too light (−6 %).



Errors on spheres, radius against time. Top: the Misner–Sharp error averaged over each sphere (red = too heavy, blue = too light). Middle: rms over the sphere of $\alpha - \alpha_{ref}$. Bottom: rms of Θ. Evaluated at t = 0 and every window end, 48 directions × 100 radii.

5. What is left: an outgoing mass deficit

The maps separate two errors. Inside the outgoing front the hole's mass is what the penalties fix: steadily too heavy without them (+27 %), right with the direct term, uniformly too light with the Bianchi term. The second error is common to all three runs: a deficit band leaves the hole at t ≈ 2 and moves out along the path of the gauge pulse, reaching −18 to −21 % at r ≈ 10–12 by t = 10. This matches the 10–19 Oct observation (“up to 17 % at r ≈ 4–12 where the gauge pulses pass”). The lapse error and Θ follow the same outgoing paths. Along a single axis the deficit can be twice its sphere average (−36 % on the ±x rays of the direct-penalty run), so it is not spherical. Where the pulse travels (r ≈ 5–15) every run rests on the sparse uniform sample, since the core and focus clouds sit at the hole, so that region is the first candidate. The control run (final settings, no penalty, running) will show whether the final settings alone change it.

6. The puncture drift: sampling and constraints

A single puncture should stay at the origin; in the defaults run it reaches 0.13 by t = 10. With puncture tracking off the push at the origin is 10× smaller. Holding the clouds at the origin only halves it, while drawing every cloud in point-reflected pairs about its centre stops it:

t = 4, defaults	**drift**	**|β| at origin, t = 3**	**non-spherical part**
clouds follow the tracked puncture	0.020	3.5e-2	0.6
clouds held at the origin	0.010	1.8e-2	0.32
point-symmetric clouds (`--symmetric_focus`)	**0.0008**	**8.3e-4**	**0.17**

The seed is the random lopsidedness within each cloud, which tracking amplifies. The constraint penalties control the drift almost as well, *without* symmetric clouds:

puncture drift $\|x_p\|$	**t = 2**	**t = 3**	**t = 5**	**t = 7.5**	**t = 10**
no penalty, defaults	1.4e-3	1.0e-2	2.7e-2	6.0e-2	0.13
no penalty, final settings (10–19 Oct)	–	–	–	–	0.023
Θ/Z^i penalty	3.5e-4	7.7e-4	9.4e-4	5.7e-3	**1.0e-2**
Bianchi penalty	3.9e-4	6.1e-4	1.2e-3	1.7e-3	**6.8e-3**

To t = 5 both stay near 1e-3, the level of the symmetric clouds at t = 4, and at t = 10 they are 13–19× below the defaults and 2–3× below the final settings alone. Why, measured on spheres of radius 0.3 about the tracked puncture (`tools/drift_dipole.py`): the tracker moves the puncture with $-\beta(x_p)$, and the Gamma-driver feeds β from the dipole (sphere average) of $\hat{\Gamma}^i$, which CCZ4 splits exactly as $\hat{\Gamma}^i = \tilde{\Gamma}^i[\tilde{\gamma}] + 2Z^i/\chi$. Without a penalty the conformal metric grows a large dipole, $|\langle\tilde{\Gamma}\rangle| \approx 0.13$ at t = 10, that $\hat{\Gamma}^i$ mostly does not follow: $2Z^i/\chi \approx -\tilde{\Gamma}^i$, a constraint violation the evolution equations allow. The ~10 % that does reach $\hat{\Gamma}^i$ ($|\langle\hat{\Gamma}\rangle| \approx 1.4\text{e-}2$) drives β (3.9e-2) and the drift. With the penalties the metric's dipole is 20–40× smaller (3.5e-3 Θ/Z^i, 6e-3 Bianchi), and $\hat{\Gamma}^i$ and β 5–10× smaller. The asymmetry enters the metric through the constraint-violating mode, and the penalties close it. The control run gives the final-settings drift on the same scale. Symmetric clouds leave the mass error unchanged; drift and mass are separate problems. The penalised runs drift 10× less without symmetric clouds.

Next steps

- **Control run** (running): final settings without a penalty, to compare the hole's mass and the outgoing deficit.
- **More interior points where the pulse travels** (r ≈ 5–15), e.g. a shell cloud following the pulse, to test whether the outgoing deficit is under-sampling.
- **Combine** the direct penalty with point-symmetric clouds and take it to the binary, where both the drift and the mass of each hole matter.